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Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

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Comment on FR Doc # 2025-02305

Submitter Information

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Organization: Click Therapeutics

General Comment

See attached file(s)

Attachments

Click Therapeutics - AI Action Plan Comment

March 15, 2025

Sethuraman Panchanathan
Director
National Science Foundation
2415 Eisenhower Avenue
Alexandria, VA 22314
Attn: Faisal D'Souza, NC

Subject: Comments on Request for Information on the Development of an Artificial Intelligence (AI) Action Plan (Docket Number: 2025-02305)

Dear Director Panchanathan,

Click Therapeutics ("Click") appreciates the opportunity to provide comments on the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan. We strongly support the Administration's efforts to sustain and enhance America's leadership in AI. As a leader in the development of prescription digital therapeutics (PDTs) leveraging AI, we believe that fostering innovation in highly specialized and value-add areas of AI, such as medical technology, is crucial for maintaining a competitive edge.

Founded in New York City in 2012, Click is redefining modern medicine with patient-centric digital treatments. Operating at the intersection of biology and technology, Click combines neuroscience with the power of software, creating a new way to treat disease. We develop, validate, and commercialize software as prescription medical treatments for people with unmet medical needs. Our products are patient-facing mobile applications where the software is the treatment; they are not telemedicine. As such, Click develops its PDT products in compliance with U.S. Food and Drug Administration (FDA) regulations for software-as-a-medical-device (SaMD) therapeutics.

As the Trump Administration develops its AI Action Plan, we offer the following recommendations:

1. **Awareness and Consideration of PDT Use Cases:** We urge the Administration to recognize and consider the unique use cases of AI in PDTs, including the application of generative AI and the utilization of large language model (LLM) foundational models and agentic AI in deploying patient-facing treatments. This specific area of AI application requires tailored guidance and regulatory frameworks that are distinct from what might apply to other AI-enabled devices, such as image analysis and pattern detection software in radiology and cardiology;

2. **Maintaining the Integrity of Regulated PDTs:** A regulatory environment needs to be established with clear boundaries that prevent non-device companion, wellness, or FDA-exempt software (including exempt clinical decision support software) from expanding their indications, functionality, or marketing claims in a manner that undermines the rigor and clinical validation of regulated PDTs, which are designed and indicated to treat disease (i.e., improve clinical outcomes). It is essential to preserve the integrity of clinically validated medical devices to ensure the safety of the American public and to provide the trust necessary for wide-spread adoption of these technologies by clinicians. If implemented in a commonsense manner, clear regulatory boundaries will allow both categories – non-device and device – to thrive; and

3. **Regulatory Flexibility for AI Innovation in PDTs:** While we believe regulation of AI technologies for medical uses will ultimately foster American leadership in AI, it must be fit-for-purpose and commonsense. For regulated medical device software, the Administration should implement added flexibility to enable the innovative use of AI without burdensome on-market requirements. This would include modifying change control requirements and the criteria for when new pre-market filings are required. To ensure patients can benefit from the full value of AI in regulated software, a regulatory system that allows for advanced AI features such as personalizing the user experience or treatment plan based on individual patient needs, is paramount. This flexibility should be predicated on a risk-based approach with appropriate controls and safeguards and should apply both to regulated SaMD as well as software for use in combination with drugs (e.g., prescription drug use-related software).

We offer greater detail on these recommendations and additional policy considerations below.

Supporting American AI Health Industry Competitiveness

In general, in the context of AI-driven health solutions, we believe that commonsense regulation can catalyze innovation and American competitiveness in the global economy. In the context of health AI, by clearly defining the boundaries between, on the one hand, generalist, wellness and administrative AI products and, on the other hand, regulated medical devices like PDTs, this Administration can create distinct categories that encourage focused innovation. This delineation allows for the development of two thriving sectors: one focused on rapid iteration and broad accessibility in non-regulated categories, and the other on rigorous evidence-based clinical validation for safety and effectiveness in medical devices tailored to specific patient populations. This clarity enables U.S. companies to compete globally in both arenas, leveraging the strengths of each category.

Moreover, patient and physician trust is paramount for the successful adoption of any medical solution, particularly those incorporating AI. Premarket validation of clinical safety and effectiveness by companies for FDA clearance/approval, as is already required for most novel regulated PDTs, is essential to establish and maintain that trust. This regulatory oversight both safeguards patient well-being and provides crucial credibility to enhance the marketability and global acceptance of U.S.-developed PDTs. In particular, FDA clearance/approval is often either a requirement or can accelerate entry into foreign markets for products that are regulated as medical devices in those jurisdictions. In this way, thoughtful regulation acts as an enabler, fostering a robust and trustworthy ecosystem for AI innovation in healthcare.

Commonsense Regulatory Approach

We believe that achieving the necessary levels of innovation and competitiveness in med tech requires a balanced regulatory approach. This approach should strike a balance between ensuring adequate assurance of safety and effectiveness and fostering agile regulatory policies that support rapid innovation.

Specifically, Click encourages the Administration to:

- Maintain premarket review requirements for AI-based technologies meeting the definition of a medical device, including requiring clinical validation of novel therapeutic products to establish a baseline for each product's safety and effectiveness, to ensure a threshold of product quality and to protect patient safety as conditions for marketing; *and*
- Enforce compliance requirements for AI-based products with a medical device intended use, such that non-validated non-device products do not 'drift' into regulated indications. This may require clear labeling requirements and/or product design guardrails that clearly delineate validated medical device AIs from non-validated wellness or administrative AIs (this may require provisions beyond FDA's current authority and/or enforcement by other Departments or Agencies). This will ensure two distinct product categories are maintained - medical device and non-device AI - which is essential for securing patient and provider trust and advancing U.S. competitiveness in medical AI globally.

We believe that establishing clear regulatory boundaries between medical device and non-device AI will provide consistency for patients, providers and businesses, and creation of a level-playing field within each category, allowing for rapid innovation within each product type.

For maximal impact, this regulatory clarity should be combined with greater flexibility for medical device AI to encourage the development of cutting-edge AI-powered PDTs. This can be achieved with:

- In the near-term, a broader interpretation of the Predetermined Change Control Plan (PCCP) to facilitate iterative improvements and adaptations of AI-driven PDTs;
- Increased flexibility in change control and prior authorization processes for AI-driven modifications, provided that these changes adhere to predefined specifications and risk mitigation strategies; and
- The ability to make AI-driven changes that may impact therapeutic safety and effectiveness without prior authorization (in contrast to existing requirements for new marketing submissions). New, innovative controls for higher risk changes, such as real-world canary testing prior to full roll-out and/or *in silico* model-based testing, could be established to enable these reforms.

We believe it is a worthwhile area for regulatory reform to achieve these changes within FDA policy, including any enabling legislation, so long as these reforms retain the right-sized FDA regulation and premarket validation requirements for PDTs that are essential for the category to remain competitive, a focus of innovation, and an area of U.S. leadership in AI.

Conclusion

Click is committed to working with the Administration to develop an AI Action Plan that fosters innovation, ensures patient safety, and strengthens America's position as a global leader in AI-driven medical technology. We believe that by focusing on specialized areas like PDTs and adopting a balanced, commonsense regulatory approach, the U.S. can fully realize the transformative potential of AI in healthcare.

Thank you for your consideration of these comments.

Sincerely,

Click Therapeutics